

BUSINESS REQUIREMENTS DOCUMENT (BRD)

Centralized Cattle Feed Management System

Client: Rakesh Kataria & Gopal (Partners)

Companies: 4 with GST Registration

Location: India

Document Date: 28th August 2026

Project Duration: 6 Weeks

Status: For Client Validation

1. EXECUTIVE SUMMARY

This BRD outlines the complete requirements for a centralized cattle feed management system that consolidates operations across 4 companies into a single unified platform.

Key Objectives:

- Centralized reporting for all 4 companies in a single dashboard
- Web-based master data entry and reporting interface
- Mobile app for customer ordering
- Comprehensive financial tracking (purchases, sales, payments, GST)
- Multi-bank account management
- 40+ detailed reports for CA/financial analysis
- 1 year free bug fixes and support included

Deliverables:

- Centralized Reporting Website (React.js)
- Mobile App for Customers (React Native)
- 40+ Comprehensive Reports
- Multi-firm Management Dashboard
- User Training & Documentation
- 24/7 Support for First Month

2. CLIENT INFORMATION

Client Partners: Rakesh Kataria & Gopal

Company Structure:

- Number of Companies: 4 different companies
- GST Registration: Each company has unique GST registration
- Business Sector: Cattle Feed Manufacturing & Distribution
- Location: India (GST compliance required)

Business Overview: The client operates a cattle feed business with 4 registered companies. They purchase cattle feed products from 20 different vendors (each supplier provides different quality grades), resulting in ~80 unique product SKUs. Products include items like "Khal" and other cattle feed ingredients with varying quality standards from different suppliers.

3. PRODUCTS & VENDOR STRUCTURE

Product Breakdown:

- 20 Unique Products (e.g., Khal, etc.)
- Multiple Quality Grades per Product
- ~80 Total Product SKUs (20 products × multiple suppliers)
- Sourced from 20 Different Vendors

Product Example:

- **Product:** Khal
- **Suppliers:** 20 vendors
- **Quality Variations:** High, Medium, Low from each vendor
- **Result:** ~20 unique SKUs for single product

Vendor Management:

- Each product purchased from multiple vendors
- Each vendor supplies different quality grades
- Fixed brokerage per vendor (paid monthly)
- Credit terms: 7, 15, or 30 days

- Quality tracking by vendor
- Price variation based on vendor and quality

4. USER PERSONAS & ROLES

Five Key Personas:

1. CEO/Owner

- Strategic oversight of all 4 companies
- Financial approvals (high-value transactions)
- User management
- View all reports and consolidated dashboards

2. Manager

- Day-to-day operations for assigned firm
- Create and approve purchase orders
- Create and approve sales orders
- Approve payments (up to limit)
- Firm-specific reporting

3. Sales Representative

- Create customer sales orders
- Process customer orders from mobile app
- Handle returns and adjustments
- Customer support and follow-ups
- View own commission and sales

4. Master Data Entry Operator

- Product master maintenance
- Vendor/supplier master maintenance
- Customer master maintenance
- Bulk data imports/uploads via Excel templates
- Data validation and corrections

5. Employee (Future Phase)

- Profile management
- Request leaves (Phase 2)
- View payslips (Phase 2)

Detailed permissions defined in separate permission matrix to be finalized during development.

5. PURCHASE ORDER PROCESS

Commodity-Based Pricing:

- Prices vary daily (commodity products)
- Rate locked at time of PO creation
- Example: Purchase 1000 bags @ ₹100/bag today
- After 10 days, purchase 500 bags @ ₹95/bag (new rate)

Partial Delivery Handling:

Scenario:

1. **Day 1:** Order 1000 bags @ ₹100/bag
2. **Day 5:** Receive 800 bags (200 pending @ ₹100)
3. **Day 15:** New order 500 bags @ ₹95/bag
4. **Day 17:** Vendor sends 700 bags:
 - 500 from new order @ ₹95/bag
 - 200 from previous pending @ ₹100/bag
5. **Result:** System tracks which bags from which PO at which rate

Cost Components (All-Inclusive):

Every PO includes:

- **Product Cost:** Quantity × Rate
- **Fixed Brokerage:** Per vendor, monthly payment
- **Vehicle Rent:** Transportation cost
- **Labour Cost:** Loading/unloading to godown
- **GST Amount:** Input GST for ITC claim
- **Total Cost Per Unit** calculated automatically

Payment Terms & Benefits:

- **Credit Terms:** 7 days, 15 days, or 30 days
- **Early Payment Discount:** 1-2% if paid within 7 days
- **Late Payment Penalty:** Extra charges if paid after 30 days

Opening Balance Handling:

- Existing POs with pending deliveries tracked separately
- Rate locked from original PO date
- No recalculation based on new rates

6. SALES ORDER & BILLING PROCESS

Dynamic Pricing:

- Rate card updated **daily at 11 AM IST**
- Customers get current market price at time of order
- Rate locked at order time, **NOT** at delivery time
- Example: Order placed at 10 AM gets rate from yesterday's 11 AM
 - Delivery after 2 days: Same rate maintained

Sales Channels:

- **Cash Sales:** Paid at time of order
- **Credit Sales:** 7 days, 15 days, or 30 days
- **Bulk/Business Clients:** Via brokers (middleman)
- **Inter-firm Sales:** Between 4 companies
- **Mobile App Orders:** Customer-initiated ordering

Discount Structure:

- **Cash/Online Payment:** 2% discount
- **Credit 7 days:** 1.5% discount
- **Credit 15 days:** 1% discount
- **Credit 30 days:** No discount (full amount)
- **Late Payment (>30 days):** Extra charges

Delivery & Fulfillment:

- Loading/unloading charges included in sales price
- Vehicle rent included in sales price
- Delivery after 2 days if stock available
- Rate locked at order time maintained at delivery

Returns & Adjustments:

- Return requests accepted
- Sales bill quantity adjusted accordingly
- Rate cannot be modified by sales person (system calculated)
- Adjustment tracked separately

Brokerage for Bulk Clients:

- Broker acts as middleman
- Monthly brokerage settlement
- Tracked and reported separately
- Fixed commission % per broker

7. MASTER DATA MANAGEMENT

Stock/Inventory:

- **One-time manual entry** of existing stock
- Quantity per product at go-live
- Auto-updates via PO and sales transactions
- No further manual updates needed

Vendor Master:

- **One-time entry** via Excel template
- **Bulk upload capability**
- Includes:
 - Vendor details
 - Bank information

- Brokerage rates
- Payment terms
- Contact information

Customer Master:

- **Created** by Master Data Entry operator
- **Edited** by Manager/Sales Rep (contact info only)
- Includes:
 - Customer details
 - Credit limit
 - Delivery addresses
 - Payment preferences

Employee Master (Future):

- Employee information entry
- **Phase 2:** Attendance tracking
- **Phase 2:** Salary management

8. GST & BANKING REQUIREMENTS

GST Handling:

- Some products have GST, some are GST-free
- GST calculated on **purchase orders**
- GST calculated on **sales invoices**
- **Input GST tracking** (ITC claim)
- **Output GST tracking** (liability to government)
- **Reports for CA:** Monthly GST reconciliation

Multi-Bank Account Management:

- Each firm has **multiple bank accounts**
- Track which account **receives customer payments**
- Track which account **pays supplier bills**
- **Bank reconciliation module** included

- **Multi-account cash flow** tracking
- Each account identified separately in reports

9. REPORTING & ANALYTICS

40+ Comprehensive Reports:

Purchase Reports (8):

- PO Summary Report
- Vendor-wise Purchase Analysis
- PO Tracking (Detailed with pending quantities)
- Purchase Price Variance
- Cost Breakdown (per PO, per unit)
- Brokerage Tracking (monthly)
- GST Report (Input Tax)
- Payables Aging

Sales Reports (10):

- Sales Order Summary
- Customer-wise Sales Analysis
- Sales Price Realization
- Sales vs Purchase Price (Profit/Loss analysis)
- Delivery Tracking
- Customer Payment Status
- Sales Returns & Adjustments
- Brokerage to Brokers (monthly)
- GST Report (Output Tax)
- Sales from Multiple Channels

Inventory Reports (4):

- Stock Summary
- Stock Movement
- Stock Valuation
- Stock Aging

Financial Reports (8):

- Profit & Loss Statement
- GST Reconciliation
- Bank Reconciliation
- Cash Flow Statement
- Accounts Receivable Aging
- Accounts Payable Aging
- Monthly Commission/Brokerage
- Opening Balances Status

Dashboards (6):

- Executive Dashboard (all 4 firms)
- Firm-wise Dashboard
- Inventory Dashboard
- Financial Health Dashboard
- Sales Performance Dashboard
- Purchase Performance Dashboard

Compliance & Audit (4):

- Transaction Audit Trail
- GST Compliance Report
- Bank Reconciliation Audit
- Opening Balance Tracking

Report Features:

- **Firm-wise reports** (individual company view)
- **Master dashboard** (all 4 firms in single view)
- **CA-audit ready** (every entry traceable)
- **Export to PDF and Excel**
- **Scheduled report delivery** via email
- **Drill-down capability** for detailed analysis

10. OPENING BALANCES AT GO-LIVE

Strategy: Opening Balance Entry via Excel Template

What to Capture:

- All outstanding customer receivables
- All outstanding vendor payables
- Existing opening stock

Implementation:

- **Pre-go-live:** Collect opening balance data from client
- **Week 5:** Client fills Excel template
- **Week 5:** Data validation and upload
- **Week 6:** Opening balance entries created in system
- **Post-go-live:** Track collections/payments separately

Accounting Treatment:

- Opening balances separated from new transactions
- Track which opening balances are collected/paid
- Monthly reconciliation reporting
- Aging analysis (how old the balance is)

11. TECHNOLOGY STACK & ARCHITECTURE

Backend:

- **Runtime:** Node.js v18 LTS
- **Framework:** Express.js
- **Database:** PostgreSQL 15+ (NOT MongoDB)
- **Cache:** Redis
- **ORM:** Prisma

Frontend (Website):

- **Framework:** React 18+
- **State Management:** Redux Toolkit
- **UI Library:** Ant Design (enterprise components)

- **Build Tool:** Vite
- **Charts:** Recharts / ECharts

Mobile App:

- **Framework:** React Native 0.72+
- **UI:** React Native Paper
- **Navigation:** React Navigation
- **State:** Redux Toolkit

Hosting:

- **Cloud:** AWS (Mumbai region - ap-south-1)
- **Compute:** EC2 with auto-scaling
- **Database:** RDS PostgreSQL (Multi-AZ)
- **Cache:** ElastiCache (Redis)
- **Storage:** S3
- **CDN:** CloudFront

Why PostgreSQL (Not MongoDB)?

- **ACID Transactions:** Critical for financial accuracy
- **Strong Consistency:** No data loss in accounting
- **Complex Queries:** Better for reporting
- **Referential Integrity:** Foreign keys support
- **Banking Standard:** Used by financial institutions
- **GST Compliance:** Better audit trail support

12. SECURITY & COMPLIANCE

Data Security:

- **Encryption at rest:** AES-256 (database + storage)
- **Encryption in transit:** TLS 1.2+ (HTTPS)
- **Data masking:** Sensitive data masked in logs
- **Backup encryption:** Daily automated encrypted backups

Authentication & Access Control:

- **JWT authentication** (24-hour tokens)
- **Role-based access control** (RBAC)
- **Multi-factor authentication** (optional)
- **Session management** (15-30 min timeout)

Compliance:

- **GST Compliance:** GSTR reports generation
- **Audit Trail:** 3-year transaction logging
- **CA-ready:** Every entry traceable for audits
- **Incident Response:** Security protocols in place

13. IMPLEMENTATION TIMELINE (6 WEEKS)

Aggressive 6-Week Delivery Schedule:

- **Week 1:** Infrastructure setup + Core API development (parallel)
- **Week 2-3:** Parallel development (Website, Mobile, Backend simultaneously)
- **Week 4:** Feature completion + Integration testing
- **Week 5:** UAT + Bug fixes + Performance optimization
- **Week 6:** Production deployment + Go-live + Stabilization

Team Size (23-24 people):

- **Backend:** 4 developers
- **Frontend:** 4 developers
- **Mobile:** 3 developers
- **QA:** 4 engineers
- **DevOps:** 2 engineers
- **Project Manager + Business Analysts:** 3 people

Key Milestones:

- **Day 7:** Core APIs ready
- **Day 21:** Website and Mobile apps beta ready

- **Day 28:** All features developed
- **Day 35:** UAT completed
- **Day 42:** Production deployment

14. DELIVERABLES

By End of Week 6:

✓ Centralized Reporting Website (React) ✓ Mobile App (React Native) - iOS & Android ✓ 40+ Reports (firm-wise + consolidated) ✓ Master Data Management Interface ✓ Purchase Order Module ✓ Sales Order Module ✓ Payment Management ✓ Stock Management ✓ GST Tracking & Reporting ✓ Multi-bank Account Management ✓ Opening Balances Integration ✓ User Training & Documentation ✓ 1 Year Free Support & Bug Fixes ✓ 24/7 Support for First Month

15. POST-GO-LIVE SUPPORT

Included Support Package:

- **Month 1:** 24/7 intensive support
- **Months 2-3:** Scheduled support + ongoing optimization
- **Year 1:** Free bug fixes and critical issue resolution
- **Support includes:** Issue resolution, performance tuning, data corrections

Enhancement Services (Additional Charges):

- New features beyond original scope
- Phase 2 modules (Attendance, Payroll)
- Third-party integrations (Banks, APIs)
- Performance optimization

16. ASSUMPTIONS & CONSTRAINTS

Assumptions:

- Client will provide timely feedback and approvals

- Opening balance data will be ready by Week 5
- Internet connectivity stable for cloud-based system
- Users trained adequately before go-live
- Browser: Chrome/Firefox latest versions

Constraints:

- 6-week timeline is fixed
- Scope must remain within defined requirements
- Major changes require separate quote
- Third-party integrations may require additional time

17. DOCUMENT APPROVAL

This BRD is prepared for validation with the client.

Client Sign-off Required:

- Rakesh Kataria (Partner)
- Gopal (Partner)

Once approved, development will commence immediately.

Document Version: 1.0

Date: 28th August 2026

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